

Youssef Elebiary

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Summary

Applied Science Intern at Microsoft and Computer Science student specializing in Retrieval-Augmented Generation (RAG), multimodal data fusion, and NLP. Proven experience designing and optimizing enterprise-grade machine learning pipelines, balancing algorithmic complexity with inference latency. Combines deep theoretical knowledge of Deep Learning and Graph Neural Networks (GNNs) with robust Python and C++ engineering to build scalable, production-ready AI systems.

Experience

Applied Science Intern, Microsoft, Cairo Jul 2026 – Aug 2026

- Engineered and optimized end-to-end **Retrieval-Augmented Generation (RAG)** pipelines, balancing algorithmic complexity against system latency and retrieval performance.
- Benchmarked and optimized AI model workflows, balancing algorithmic complexity against inference latency, computational cost, and system efficiency.
- Collaborated with senior applied scientists and engineers within an enterprise framework to build, evaluate, and integrate scalable machine learning components.

Information Technology Intern, FPi-Future Pharmaceutical industries, Cairo Aug 2025 – Sep 2025

- Enhanced a custom task management system by adding a "comment" feature that improved teamwork.
- Contributed to an Order Receiving mobile app (Kotlin + Django API), partially integrated with the JD Edwards (JDE) ERP system.
- Applied back-end development best practices while gaining hands-on experience with .NET and Angular.

NLP Engineer Intern, Elevvo Pathways, Cairo Jul 2025 – Aug 2025

- Fine-tuned a **BART-base** model for abstractive text summarization, achieving a **ROUGE-1 score of 39.9%** on the CNN-DailyMail benchmark.
- Built end-to-end classification pipelines for **Sentiment Analysis** and **Fake News Detection** using TF-IDF and optimized standard classifiers.
- Implemented unsupervised topic modeling using **LDA** and **NMF**, achieving a superior **coherence score of 0.77** with NMF to extract latent themes from unstructured news data.

Education

BSc in Computer Science, **Cairo University** Sep 2023 – Jul 2027

- GPA: 3.53

Certifications

Microsoft Machine Learning Engineer , DEPI	Dec 2025
IBM AI Engineering Specialization , IBM – Coursera	Sep 2024
HCIA – Artificial Intelligence , Huawei ICT Academy	Sep 2024
Object Oriented Programming Specialization , University of London – Coursera	Sep 2024
Google AI Essentials , Google – Coursera	Sep 2024

Projects

RescuNet - AI Disaster Response Platform

github.com/YoussefElebiary/RescuNet

- Developed a **Graph Neural Network (GINEConv)** to predict road traversability on highly imbalanced, sparse graphs and engineered a **C++ routing engine** to solve VRP in milliseconds.
- Integrated **YOLOv11** models for RGB hazard/human detection and Thermal human detection, alongside an **NLP LSTM** model to classify and prioritize distress signals.
- Built a full **React 19** website with **MapLibre** and **FastAPI** enabling users to visualize live drone feeds, **perform real-time text analysis**, and generate dynamic rescue routes.

Autonoma - Autonomous ML Pipeline

github.com/YoussefElebiary/Autonoma

- Automated the end-to-end machine learning lifecycle (EDA, preprocessing, and training) by engineering a **LangGraph** orchestration system that dynamically routes distilled state context, preventing local LLM VRAM exhaustion.
- Decoupled AI reasoning from code execution by integrating the **Model Context Protocol (FastMCP)**, securely executing generated JSON tool calls via **Polars** and **Scikit-Learn**.
- Designed a self-correcting Critic agent loop to autonomously audit preprocessing and modeling steps for data leakage and statistical fallacies prior to training and tuning predictive models.

AI Image Captioning

github.com/YoussefElebiary/VisionScribe

- Developed a captioning architecture using **InceptionV3** for feature extraction and **Bidirectional LSTMs** for sequence generation.
- Achieved a **BLEU-1 score of 52.96%** and **BLEU-4 of 15.34%**, validating the model's ability to generate semantically accurate descriptions.
- Implemented **Beam Search** decoding to improve caption quality and deployed the system via a **Flask** REST API with a web interface.

Fake News Classifier

github.com/YoussefElebiary/TruthGuard

- Engineered a dual-model classification system using **XGBoost** and **LightGBM**, achieving a **99.8% F1-Score** and **99.9% Precision** on the validation set.
- Developed a **Flask** web interface that allows users to toggle between high-precision and lightweight inference modes.

Image Editing Software

github.com/YoussefElebiary/PhotoWizard

- Built a desktop GUI app in **C++** using **Qt** to apply visual filters to images with real-time previews.
- Supported multiple image formats and optimized performance for smooth user interaction.

Skills

AI & Machine Learning

PyTorch, TensorFlow/Keras
Scikit-learn, XGBoost
NLP (Transformers, LLMs)
Computer Vision, GNNs

Tools & Languages

Git, LangGraph, Qt
REST APIs, Polars
English (C1/Fluent)
Arabic (Native)

Software Engineering

Python, C++
Flask, Django, FastAPI
React, HTML/CSS
Data Structures & Algorithms

Extracurricular Activities

MSP Tech Club - Helwan University

Head of AI & Data Science

Jan 2026 – Present

- Leading the AI committee to design technical workshops and mentor junior members.
- Orchestrating the season plan to align with current industry trends in Data Science.

AI Committee Member

May 2025 – Dec 2025

- Collaborated on internal AI projects and organized community tech initiatives.